

ADVANCED DATA STRUCTURES
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Name the algorithm used for finding transitive closure of a directed graph.	M 1.01	R
2	State the use of Dijkstra's algorithm.	M 1.03	R
3	Define a minimum spanning tree.	M 1.04	R
4	_____ is the primary operation to rebalance a binary search tree.	M 2.02	R
5	State the AVL tree property.	M 2.03	R
6	Define collision in hashing.	M 3.04	R
7	List two operations on sets.	M 3.05	R
8	Write the names of two design techniques employed by text processing algorithms.	M 4.01	R
9	Information about web pages in the world wide web are gathered by a program called _____	M 4.06	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	Define transitive closure of a directed graph. Give an example.	M 1.01	R
2	State three applications of Directed Acyclic Graphs.	M 1.02	R
3	List the rotation operations in a binary search tree. Give examples for each.	M 2.02	R

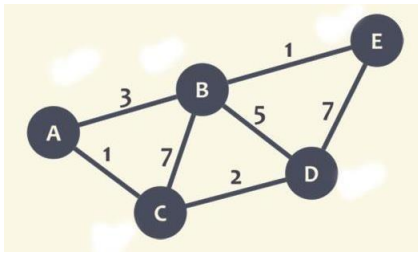
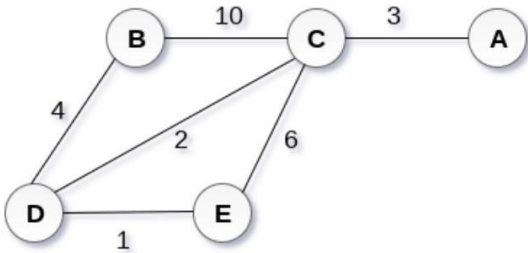
4	Define AVL Tree. Give an example.	M 2.03	R
5	Describe balanced search trees with an example.	M 2.01	R
6	State any three operations which can be performed on the Map ADT.	M 3.01	R
7	State division hash function. Give an example.	M 3.03	R
8	Describe linear probing for collision resolution.	M 3.04	R
9	Define set, multiset and multimap.	M 3.01	R
10	Give three examples of digital collections that include textual information.	M 4.01	R

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Using Dijkstra's algorithm, find the shortest path from the source node C to all other nodes in the following graph. 	M 1.03	A
OR			
2	Find the Minimum Spanning Tree of the following graph. Use Prim's algorithm. 	M 1.04	A
3	Write the Floyd Warshall algorithm for transitive closure.	M 1.01	U
OR			
4	Write the Kruskal's algorithm for Minimum Spanning Tree.	M 1.05	U

5	Illustrate the insertion of a node in an AVL Tree.	M 2.05	U
OR			
6	Illustrate the deletion of a node in an AVL Tree.	M 2.06	U
7	Explain set ADT and its operations.	M 3.05	U
OR			
8	Explain the structure of hash tables with an example.	M 3.02	U
9	Write the brute force pattern matching algorithm.	M 4.02	U
OR			
10	Write the Huffman coding algorithm for text compression.	M 4.03	U
11	Explain the categories of tries.	M 4.05	U
OR			
12	Draw the frequency array and Huffman tree for the following string: "dogs do not spot hot pots or cats".	M 4.03	A

